indicators are hardwired to the cluster and some indicators are controlled by messages received on the PCI Bus. The vehicle Chime function, Courtesy Lamp Output, the Battery Saver function, and the Rear Window Defogger timer are contained internally within the cluster. The cluster contains a vacuum fluorescent (VF) display for the Odometer/ Trip function. The cluster VF will also display P-codes and vehicle status messages. The cluster has the ability to store DTCs, communicate on the PCI Bus, display engine information, and display certain inputs using the DRBIII®. The cluster is also able to perform a manual self-test. For complete description and operation of the Instrument Cluster, refer to the TJ Service Manual Instrument Cluster section. For diagnostic procedures, refer to the TJ Body Diagnostic Procedures Manual.

3.5.2 INSTRUMENT CLUSTER SELF TEST

The Instrument Cluster is capable of performing a diagnostic self test. The self test is actuated by depressing and holding the trip reset button while turning the ignition from the off to the on position. The self test can also be initiated using the DRBIII®. The self test will terminate if the tachometer is greater than 300 RPM, the ignition is turned off, or the test is complete. For diagnostic procedures, refer to the TJ Body Diagnostic Procedures Manual.

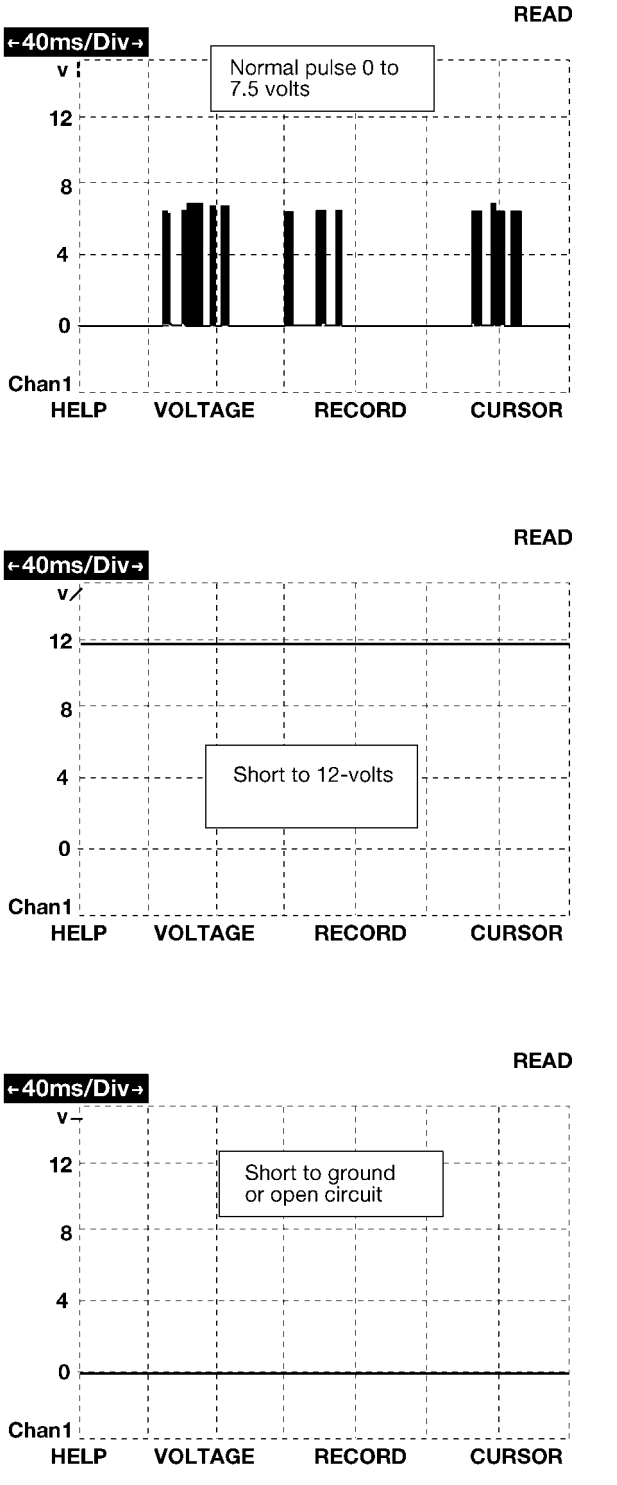
3.6 VEHICLE COMMUNICATION

The Programmable Communication Interface or PCI Bus is a single wire multiplexed network capable of supporting binary encoded messages shared between multiple modules. The PCI bus circuit is identified as D25 and is white with a violet tracer. The modules are wired in parallel. Connections are made in the harness using splices. The following modules are used on 2006 Jeep® Wrangler (TJ).

- Airbag Control Module
- Mechanical Instrument Cluster
- Radio
- Powertrain Control Module (NGC)
- Sentry Key Immobilizer Module
- Satellite Radio Receiver
- Controller Antilock Brake (Not on the PCI Bus. Communication with the scan tool is established thru the SCI Transmit circuit).

Each module provides its own bias and termination in order to transmit and receive messages. The bus voltage is at zero volts when no modules are transmitting and is pulled up to about seven and a half volts when modules are transmitting.

The bus messages are transmitted at a rate averaging 10800 bits per second. Since there is only voltage present when the modules transmit and the message length is only about 500 milliseconds, it is ineffective to try and measure the bus activity with a conventional voltmeter. The preferred method is to use the DRBIII® lab scope. The 12v square wave selection on the 20-volt scale provides a good view of the bus activity. Voltage on the bus should pulse between zero and about seven and a half volts. Refer to the following figure for some typical displays.



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